

hyper2kvm Multi-Distro Migration

4 VMware VMs → KVM on Bare-Metal AlmaLinux 9

Live end-to-end migration demonstration: Rocky Linux 9.4, Ubuntu 24.04 LTS, Ubuntu Server 22.04 LTS, and CentOS Stream 9 — all converted from VMware VMDK to KVM with automated offline fixes, virtio driver injection, netplan repair, and verified SSH connectivity. Single command per VM. Zero manual intervention.

4 Distros | VMware VMDK → KVM QCOW2 | VirtIO | Libvirt | DHCP + SSH Verified

Host: Worldstream Bare-Metal | AlmaLinux 9.7 | Intel Xeon E3-1230 v6 | 16 GB RAM | QEMU 9.1 + Libvirt 10.10

Test Environment: Bare-Metal Host

All migrations executed on a Worldstream dedicated server running AlmaLinux 9.7 — a RHEL-derivative without `qemu-system-x86_64` or SPICE, validating hyper2kvm's cross-distro auto-detection.

Host Hardware

```
Hostname:  NLDW3-4-31-12
Provider:  Worldstream (Netherlands)
CPU:       Intel Xeon E3-1230 v6 @ 3.50GHz
           4 cores / 8 threads
RAM:       16 GB DDR4
Disk:      219 GB (SSD)
Network:   1 Gbps public (185.165.240.5)
```

Host Software Stack

```
OS:         AlmaLinux 9.7 (Moss Jungle Cat)
Kernel:     5.14.0-611.41.1.el9_7.x86_64
QEMU:       9.1.0 (/usr/libexec/qemu-kvm)
Libvirt:    10.10.0
Python:     3.11.13
h2kvmctl:   0.3.0
Backend:    VMCraft (pure Python + qemu-nbd)
Graphics:   VNC (auto-detected, no SPICE)
```

Cross-Distro Challenges Solved

QEMU Binary Path

AlmaLinux uses `/usr/libexec/qemu-kvm` instead of Fedora's `/usr/bin/qemu-system-x86_64`. hyper2kvm auto-detects the correct binary across Fedora, RHEL, AlmaLinux, Debian, and Ubuntu.

Graphics Backend

AlmaLinux QEMU is built without SPICE support. hyper2kvm detects this and falls back to VNC automatically — no user configuration needed.

Netplan VMware NIC Names

Ubuntu VMs use `ens33` (VMware PCI slot). On KVM it becomes `ens3`. hyper2kvm replaces hardcoded names with `match: name: "en*" for universal compatibility.`

Initramfs VirtIO Injection

VMware VMs ship without virtio drivers in initramfs. hyper2kvm runs `dracut --add-drivers virtio_blk virtio_net...` inside the guest via chroot to ensure first-boot success.

Demo script: `run-demo.sh <vmdk-path>` — single command handles cleanup, conversion, offline fixes, libvirt define, VM boot, and IP verification.

1. Rocky Linux 9.4 (Blue Onyx)

RHEL-compatible enterprise Linux — LVM on XFS, UEFI boot, ifcfg networking.

Source

```
Source:   RockyLinux_9.4_VMM_LinuxVMImages.COM.vmdk (1.88 GiB)
Format:   VMware monolithicSparse VMDK
Layout:   GPT → /boot/efi (600M) + /boot (1G) + LVM (rl_192: root 70G + swap 2G + home 438G)
Kernel:   5.14.0-427.13.1.el9_4.x86_64
```

Migration Command

```
$ run-demo.sh /root/hyper2kvm/RockyLinux_9.4_VMM_LinuxVMImages.COM.vmdk
```

Pipeline Output (key steps)

```
✓ Sanity checks: 5/5 passed
✓ VMDK inspection: monolithicSparse, GPT, UEFI detected
✓ qemu-img convert: VMDK → QCOW2 (virtual: 512 GiB, actual: 2.1 GiB)
✓ LVM activated: VG rl_192 (root + swap + home)
✓ Offline fixes: fstab stabilized, VMware tools masked
✓ Initramfs rebuilt: dracut --add-drivers virtio_blk virtio_scsi virtio_net nvme ahci --add lvm dm
✓ Domain XML emitted: UEFI + Q35 + VirtIO + VNC
✓ virsh define + start: RUNNING
```

Result

virsh dominfo

```
Name:      RockyLinux_9.4_VMM_LinuxVMImages.COM
State:     running
CPU(s):    2
Memory:    2048 MiB
Firmware:  UEFI (OVMF)
Disk:      vda (virtio, qcow2)
Network:   virtio (virbr0)
```

SSH Verification

```
$ ssh rockylinux@192.168.122.17

rocky9.linuxvmimages.local
Rocky Linux 9.4 (Blue Onyx)
5.14.0-427.13.1.el9_4.x86_64

enp1s0  UP   192.168.122.17/24

vda (virtio, LVM: rl_192)
```

```
|rl_192-root 70G /  
|rl_192-swap 2G [SWAP]  
|rl_192-home 438G /home
```

✓ Rocky Linux 9.4 — Migration Successful

VMDK → QCOW2 | UEFI + VirtIO + LVM | IP: 192.168.122.17 | SSH accessible | ~55s total

2. Ubuntu 24.04 LTS (Noble Numbat)

Latest Ubuntu LTS with GNOME desktop — LVM on ext4, BIOS boot, netplan + NetworkManager.

Source

Source: Ubuntu_24.04_VM_LinuxVMImages.COM.vmdk (6.0 GiB)
Format: VMware monolithicSparse VMDK
Layout: MBR → /boot (1M+2G) + LVM (ubuntu-vg: ubuntu-lv 122G)
Kernel: 6.8.0-31-generic
Desktop: GNOME (snap-based Firefox, firmware-updater, snap-store)

Migration Command

```
$ run-demo.sh /root/hyper2kvm/Ubuntu_24.04_VM_LinuxVMImages.COM.vmdk
```

Pipeline Output (key steps)

- ✓ Sanity checks: 5/5 passed
- ✓ qemu-img convert: VMDK → QCOW2 (virtual: 124 GiB, actual: 6.0 GiB)
- ✓ LVM activated: VG ubuntu-vg (ubuntu-lv)
- ✓ Offline fixes: fstab stabilized, VMware tools masked
- ✓ Initramfs rebuilt: dracut --add-drivers virtio_blk virtio_net --add lvm
- ✓ Boot mode: BIOS (i440fx)
- ✓ Domain XML emitted + virsh define + start: RUNNING

Result

virsh dominfo

Name: Ubuntu_24.04_VM_LinuxVMImages.COM
State: running
CPU(s): 2
Memory: 2048 MiB
Firmware: BIOS
Disk: vda (virtio, qcow2)
Network: virtio (virbr0)

SSH Verification

```
$ ssh ubuntu@192.168.122.254

ubuntu2404
Ubuntu 24.04 LTS (Noble Numbat)
6.8.0-31-generic

ens3  UP  192.168.122.254/24

vda (virtio, LVM: ubuntu-vg)
```

```
└─ubuntu--vg-ubuntu--lv 122G /
```

✓ Ubuntu 24.04 LTS — Migration Successful

VMDK → QCOW2 | BIOS + VirtIO + LVM | IP: 192.168.122.254 | SSH accessible | ~52s total

3. Ubuntu Server 22.04 LTS (Jammy Jellyfish)

Server LTS with netplan DHCP on ens33 (VMware) — demonstrates automatic netplan NIC name fix.

Source

```
Source:   UbuntuServer_22.04_VM_LinuxVMImages.COM.vmdk (4.1 GiB)
Format:   VMware monolithicSparse VMDK
Layout:   MBR → /boot (1M+2G) + LVM (ubuntu-vg: ubuntu-lv 100G)
Kernel:   5.15.0-30-generic
Network:   netplan with hardcoded ens33 (VMware PCI slot name)
```

The Problem & The Fix

Before (VMware)

```
# /etc/netplan/00-installer-config.yaml
network:
  ethernets:
    ens33:      # ← VMware-specific name
      dhcp4: true
      version: 2
```

On KVM, the NIC becomes ens3 — netplan won't match ens33, so **no network**.

After (hyper2kvm auto-fix)

```
# /etc/netplan/00-installer-config.yaml
network:
  ethernets:
    all-en:
      dhcp4: true
      match:
        name: en*  # ← matches any backend
      version: 2
```

hyper2kvm detects VMware NIC names (ens33, ens160, etc.) and replaces with wildcard match. **Works everywhere.**

Pipeline Log

```
✓ Fixed /etc/netplan/00-installer-config.yaml (netplan) - 1 fix applied
  eth-ens33-replaced-vmware-nic-name-with-match
```

Result

virsh dominfo

SSH Verification

```
Name:      UbuntuServer_22.04_VM_LinuxVMImages.COM
State:     running
CPU(s):    2
Memory:    2048 MiB
Firmware:  BIOS
Disk:      vda (virtio, qcow2)
Network:   virtio (virbr0)
```

```
$ ssh ubuntu@192.168.122.215

ubuntuserver2204
Ubuntu 22.04 LTS (Jammy Jellyfish)
5.15.0-30-generic

ens3  UP  192.168.122.215/24

vda (virtio, LVM: ubuntu-vg)
└─ubuntu--vg-ubuntu--lv 100G /
```

✓ Ubuntu Server 22.04 — Migration Successful (with netplan auto-fix)

VMDK → QCOW2 | BIOS + VirtIO + LVM | Netplan ens33 → en* | IP: 192.168.122.215 | SSH accessible | ~50s total

4. CentOS Stream 9

RHEL upstream continuous-delivery distribution — LVM on XFS, BIOS boot, ifcfg/NetworkManager networking.

Source

```
Source:    CentOS_Stream_9_VMM_LinuxVMImages.COM.vmdk (1.4 GiB)
Format:    VMware monolithicSparse VMDK
Layout:    MBR → /boot (1G) + LVM (cs: root 70G + swap 2G + home 439G)
Kernel:    5.14.0-22.el9.x86_64
```

Migration Command

```
$ run-demo.sh /root/hyper2kvm/CentOS_Stream_9_VMM_LinuxVMImages.COM.vmdk
```

Pipeline Output (key steps)

- ✓ Sanity checks: 5/5 passed
- ✓ qemu-img convert: VMDK → QCOW2 (virtual: 512 GiB, actual: 1.4 GiB)
- ✓ LVM activated: VG cs (root + swap + home)
- ✓ Offline fixes: fstab stabilized, VMware tools masked, ifcfg MAC pinning removed
- ✓ Initramfs rebuilt: dracut --add-drivers virtio_blk virtio_scsi virtio_net nvme ahci --add lvm dm
- ✓ Domain XML emitted + virsh define + start: RUNNING

Result

virsh dominfo

```
Name:    CentOS_Stream_9_VMM_LinuxVMImages.COM
State:    running
CPU(s):    2
Memory:    2048 MiB
Firmware:    UEFI (OVMF)
Disk:    vda (virtio, qcow2)
Network:    virtio (virbr0)
```

SSH Verification

```
$ ssh centos@192.168.122.165

centosstream9.linuxvmiages.local
CentOS Stream 9
5.14.0-22.el9.x86_64

ens3  UP  192.168.122.165/24

vda (virtio, LVM: cs)
├─cs-root  70G  /
└─cs-swap  2G  [SWAP]
```

└─cs-home 439G /home

✓ CentOS Stream 9 — Migration Successful

VMDK → QCOW2 | UEFI + VirtIO + LVM | IP: 192.168.122.165 | SSH accessible | ~57s total

Summary: 4/4 Migrations Successful

All four VMware VMDKs converted and booted on KVM with a single command each. Zero manual intervention.

#	Distribution	Kernel	Boot	Storage	Network Fix	IP Address	SSH
1	Rocky Linux 9.4	5.14.0-427.13.1.el9_4	UEFI	LVM (rl_192) XFS	ifcfg MAC removed	192.168.122.17	✓
2	Ubuntu 24.04 LTS	6.8.0-31-generic	BIOS	LVM (ubuntu-vg) ext4	netplan OK	192.168.122.254	✓
3	Ubuntu Server 22.04	5.15.0-30-generic	BIOS	LVM (ubuntu-vg) ext4	netplan ens33 → en*	192.168.122.215	✓
4	CentOS Stream 9	5.14.0-22.el9	UEFI	LVM (cs) XFS	ifcfg MAC removed	192.168.122.165	✓

What hyper2kvm Did Automatically

Disk Conversion

- VMware VMDK → QCOW2
- LVM volume group activation
- Format validation

Offline Fixes

- fstab UUID stabilization
- VMware tools masked
- Network config repair

Boot Repair

- VirtIO driver injection via dracut
- GRUB config updated
- UEFI/BIOS auto-detect

Deploy

- Libvirt XML generation
- QEMU path auto-detect
- VNC/SPICE auto-detect

Host Compatibility Fixes Applied

Issue	Fedora/Debian	AlmaLinux 9	hyper2kvm Fix
QEMU binary	/usr/bin/qemu-system-x86_64	/usr/libexec/qemu-kvm	find_qemu_binary() auto-detect
Graphics	SPICE available	No SPICE (VNC only)	_default_graphics() checks libspice-server.so
Netplan NIC	N/A	N/A	Replace ens33/ens160 with match: name: "en*"
Python	3.12+	3.9 (system), 3.11 installed	Works on Python 3.10+

4/4 Distros Migrated Successfully — Zero Manual Steps

Rocky Linux 9.4 + Ubuntu 24.04 + Ubuntu Server 22.04 + CentOS Stream 9

VMware VMDK → KVM QCOW2 with VirtIO + LVM + UEFI/BIOS

All VMs booted, got DHCP IPs, and are SSH-accessible

Host: AlmaLinux 9.7 bare-metal (Worldstream) — validated cross-distro compatibility

